

Nama :Raihan Muhammadi Riswandi

Nim: 24110500001

Bagian A: Teori

Soal 1

👉 Apa perbedaan utama antara SQL dan PL/pgSQL?

- **SQL** adalah bahasa deklaratif untuk query data (SELECT, INSERT, UPDATE, DELETE). Fokus pada *apa* yang ingin dicapai.
 - **PL/pgSQL** adalah procedural language di PostgreSQL yang memungkinkan membuat function, loop, kondisi, variabel, error handling, dll. Fokus pada *bagaimana* proses dilakukan.
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Soal 2

👉 Syntax assignment yang benar dalam PL/pgSQL:

✅ b) **counter := counter + 1;**

Soal 3

👉 Urutan block PL/pgSQL yang benar:

✅ c) **DECLARE, BEGIN, EXCEPTION, END**

Soal 4

👉 Jika variable dalam DECLARE tidak diberi initial value:

- Variabel otomatis di-assign **NULL**. Jadi nilainya tidak terdefinisi sampai diberi nilai.
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Soal 5

👉 Cara yang benar untuk concatenate string:

✅ c) **first_name || ' ' || last_name**

Bagian B: Implementasi Praktis

Soal 6

-- Function 1: Hitung Weighted Grade

```
CREATE OR REPLACE FUNCTION calculate_weighted_grade(  
    midterm DECIMAL,  
    final DECIMAL,  
    assignment DECIMAL  
) RETURNS DECIMAL AS $$  
BEGIN  
    RETURN (midterm * 0.3) + (final * 0.5) + (assignment * 0.2);  
END;  
$$ LANGUAGE plpgsql;
```

-- Function 2: Validasi Email Format

```
CREATE OR REPLACE FUNCTION validate_email(email_address TEXT)  
RETURNS BOOLEAN AS $$  
BEGIN  
    -- Email dianggap valid jika mengandung '@' dan '.'  
    IF email_address ~* '^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$' THEN  
        RETURN TRUE;  
    ELSE  
        RETURN FALSE;  
    END IF;  
END;  
$$ LANGUAGE plpgsql;
```

-- Function 3: Format Student Name

```
CREATE OR REPLACE FUNCTION format_student_name(first_name TEXT,  
last_name TEXT)  
RETURNS TEXT AS $$  
BEGIN  
    RETURN INITCAP(first_name) || ' ' || INITCAP(last_name);  
END;  
$$ LANGUAGE plpgsql;
```

Bagian C: Problem Solving

Soal 7

```
CREATE OR REPLACE FUNCTION generate_student_report(student_id INT)  
RETURNS TEXT AS $$
```

```

DECLARE
    v_student RECORD;
    v_enroll RECORD;
    v_final_grade DECIMAL;
    v_status TEXT;
    v_report TEXT;
BEGIN
    -- Ambil data student
    SELECT * INTO v_student FROM students WHERE id = student_id;
    IF NOT FOUND THEN
        RETURN '✗ Error: Student not found';
    END IF;

    -- Ambil data enrollment
    SELECT * INTO v_enroll FROM enrollments WHERE student_id = student_id;
    IF NOT FOUND THEN
        RETURN '✗ Error: Enrollment not found';
    END IF;

    -- Jika ada nilai NULL, jadikan 0
    v_enroll.midterm_score := COALESCE(v_enroll.midterm_score, 0);
    v_enroll.final_score := COALESCE(v_enroll.final_score, 0);
    v_enroll.assignment_score := COALESCE(v_enroll.assignment_score, 0);

    -- Hitung final grade
    v_final_grade := calculate_weighted_grade(v_enroll.midterm_score,
    v_enroll.final_score, v_enroll.assignment_score);

    -- Tentukan status akademik
    IF v_final_grade >= 90 THEN
        v_status := 'Excellent';
    ELSIF v_final_grade >= 80 THEN
        v_status := 'Good';
    ELSIF v_final_grade >= 70 THEN
        v_status := 'Fair';
    ELSE
        v_status := 'Needs Improvement';
    END IF;

    -- Format laporan
    v_report := 'Student: ' || UPPER(v_student.first_name || ' ' || v_student.last_name)
        || ' (' || v_student.student_number || ') - Final Grade: '
        || TO_CHAR(v_final_grade, 'FM999.00') || ' - Status: ' || v_status;

```

```
RETURN v_report;

EXCEPTION
  WHEN OTHERS THEN
    RETURN '✗ Error generating report: ' || SQLERRM;
END;
$$ LANGUAGE plpgsql;
```

 Contoh penggunaan:

```
SELECT generate_student_report(1);
-- Output: Student: AHMAD SUTRISNO (2021001) - Final Grade: 85.50 - Status:
Good

SELECT generate_student_report(2);
-- Output: Student: SITI NURHALIZA (2021002) - Final Grade: 86.10 - Status: Good
```
